

Graphs and Search

Analysis of Architectures, Algorithms, and Historical Evolution

The 2026 Inflection Point

Bridging Theory and Engineering

Game artificial intelligence has reached a critical inflection point in 2026. The discipline now perfectly bridges complex mathematical graph theory with high-performance software engineering.

Driven by strict hardware constraints, integrating Generative AI workflows and Data-Oriented Design (DOD) has become absolutely paramount for executing massive, scalable simulations in modern engines like Unreal 5.6 and Unity DOTS.



Applied Graph Typologies



Navigation Graphs

Ubiquitous spatial blueprints mapping physical game worlds. Nodes connect adjacent tiles or polygons (NavMeshes), allowing AI agents to ascertain viable routes, avoid obstacles, and calculate optimal trajectories toward dynamic destinations.



Dependency Graphs

Directed Acyclic Graphs (DAGs) mapping logical prerequisites rather than space. Essential for real-time strategy games, they allow AI opponents to plan long-term strategic objectives by tracking technology trees and construction sequences.



State Graphs

Abstract representations where nodes symbolise complete puzzle configurations and edges denote valid transition actions. Utilised extensively to solve combinatorial logic puzzles, allowing AI to navigate complex possibility spaces toward goal states.

Graph Representation in Memory

Translating theoretical models into executable code dictates algorithmic complexity and CPU cache utilisation.

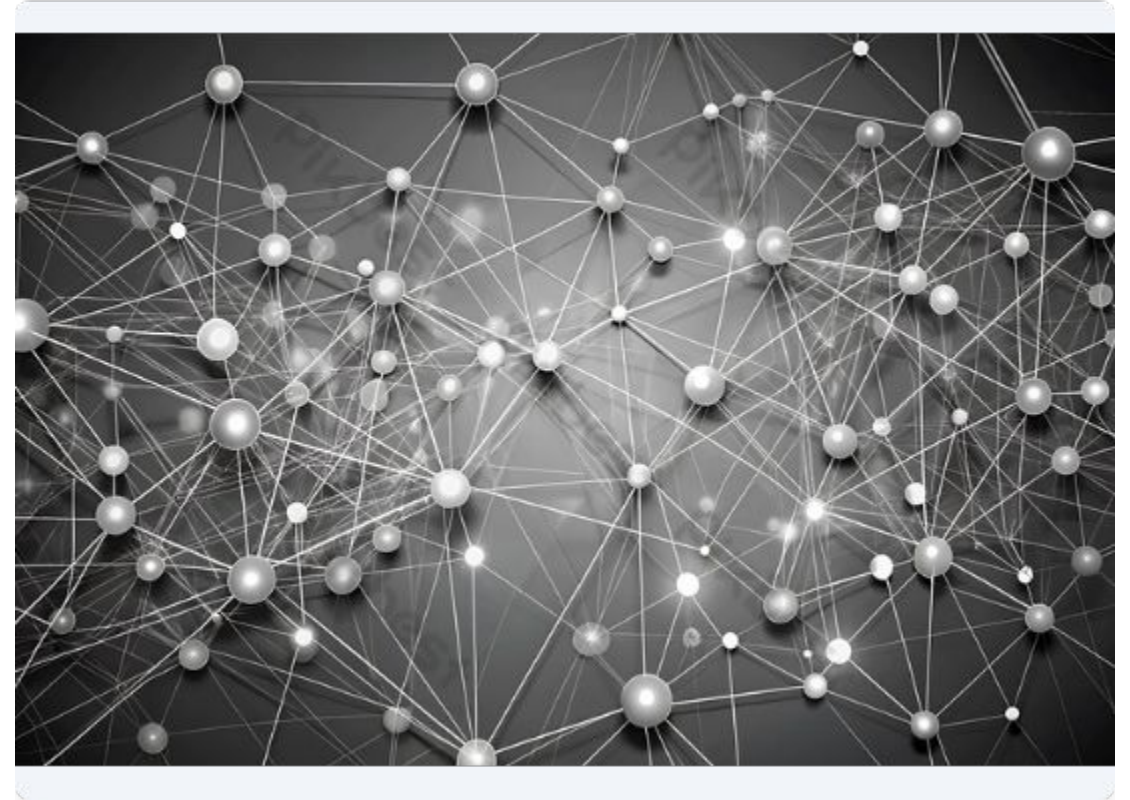
Representation Type	Space Complexity	Lookup Time (Edge)	Optimal Game Use Case
Adjacency Matrix	$O(N ^2)$	$O(1)$	Dense graphs, highly interconnected webs, and very small abstract state spaces. Prone to extreme CPU cache misses.
Adjacency List	$O(N + E)$	$O(\text{Degree}(N))$	Sparse graphs, massive spatial navigation grids, and open-world NavMeshes. The absolute industry standard.

Abstract Search: BFS Implementation

Uninformed Graph Traversal

Breadth-First Search (BFS) is a fundamental uninformed strategy. It explores graphs uniformly outward using a First-In-First-Out (FIFO) queue structure. While completely optimal for unweighted graphs, its exponential memory demands make it suitable primarily for procedural generation and specific flow fields.

```
def bfs(graph, start):  
    visited = []  
    queue = [start]  
  
    while queue:  
        node = queue.pop(0)  
        if node not in visited:  
            visited.append(node)  
            queue.extend(graph[node])  
  
    return visited
```



Search Methodologies

Uninformed (Blind) Search

Operates without any domain-specific knowledge of the problem space or distance to the goal.

Strategies like BFS and DFS rely heavily on foundational queue and stack structures. While mathematically complete, tracking the expanding search perimeter causes memory consumption to grow exponentially, rendering them inefficient for massive 3D open-world environments.

Informed (Heuristic) Search

Exploits distinct domain characteristics by introducing a heuristic distance estimator.

The A* algorithm evaluates nodes using the composite function . $f(n) = g(n) + h(n)$

By utilising geometric heuristics (like Euclidean distance), it aggressively prunes irrelevant branches, forming the highly optimised core of all modern character pathfinding.

Engine Architecture: DOD vs OOP

Performance Gain

Overcoming the "Memory Wall"

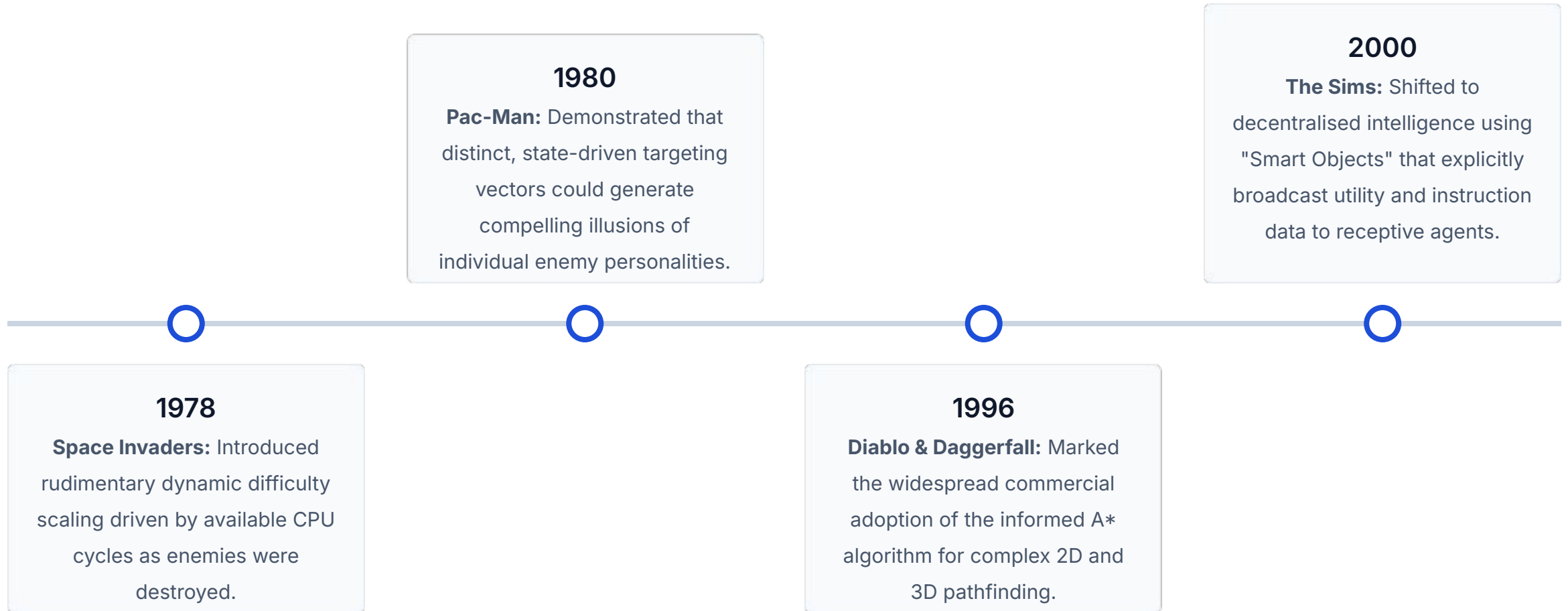
The industry has executed a systemic pivot from Object-Oriented Programming (OOP) to Data-Oriented Design (DOD) to combat severe CPU cache misses and frame rate latency.

By adopting Entity Component Systems (ECS) in Unreal Mass AI and Unity DOTS, data is strictly organised into contiguous, homogeneous arrays. This dense physical memory layout allows processors to utilise SIMD vectorisation, comfortably updating tens of thousands of complex AI agents concurrently.

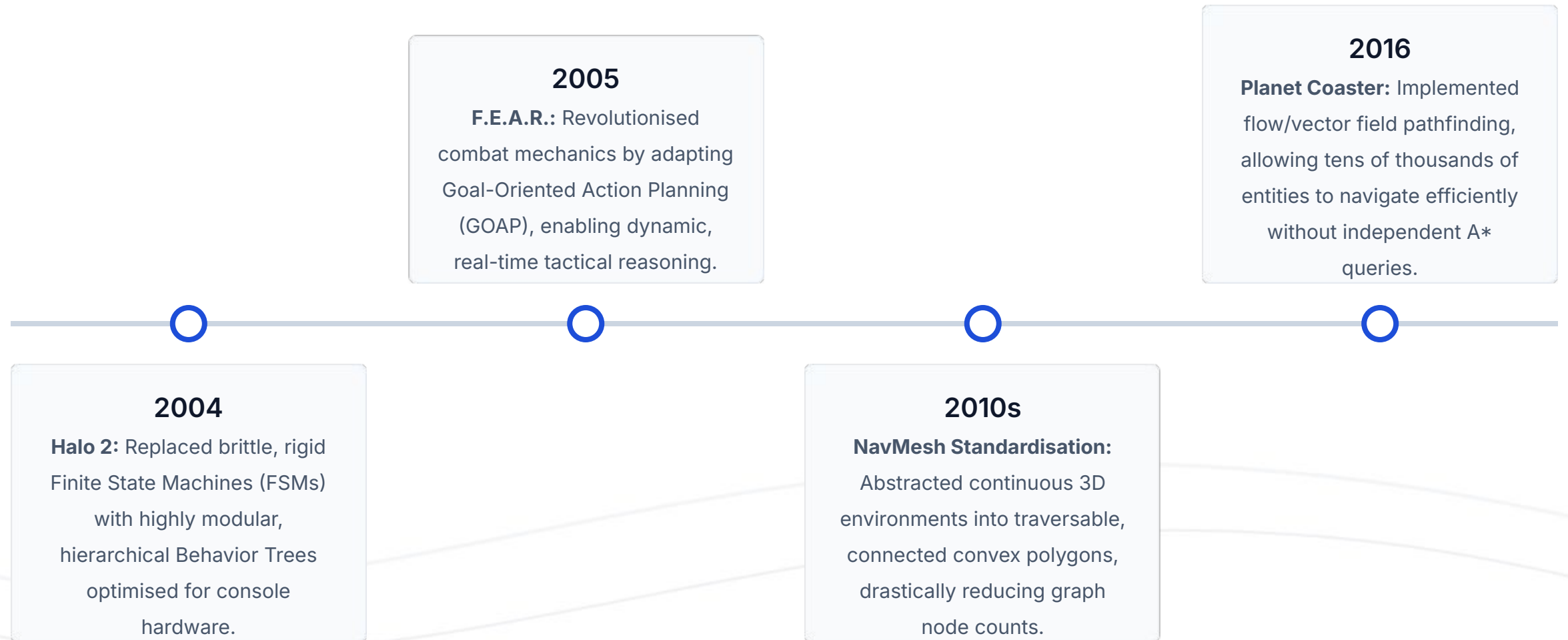
The Historical Evolution of Game AI

Tracking the structural trajectory from deterministic arcade patterns to highly parallelised neural processing units (1978–2027).

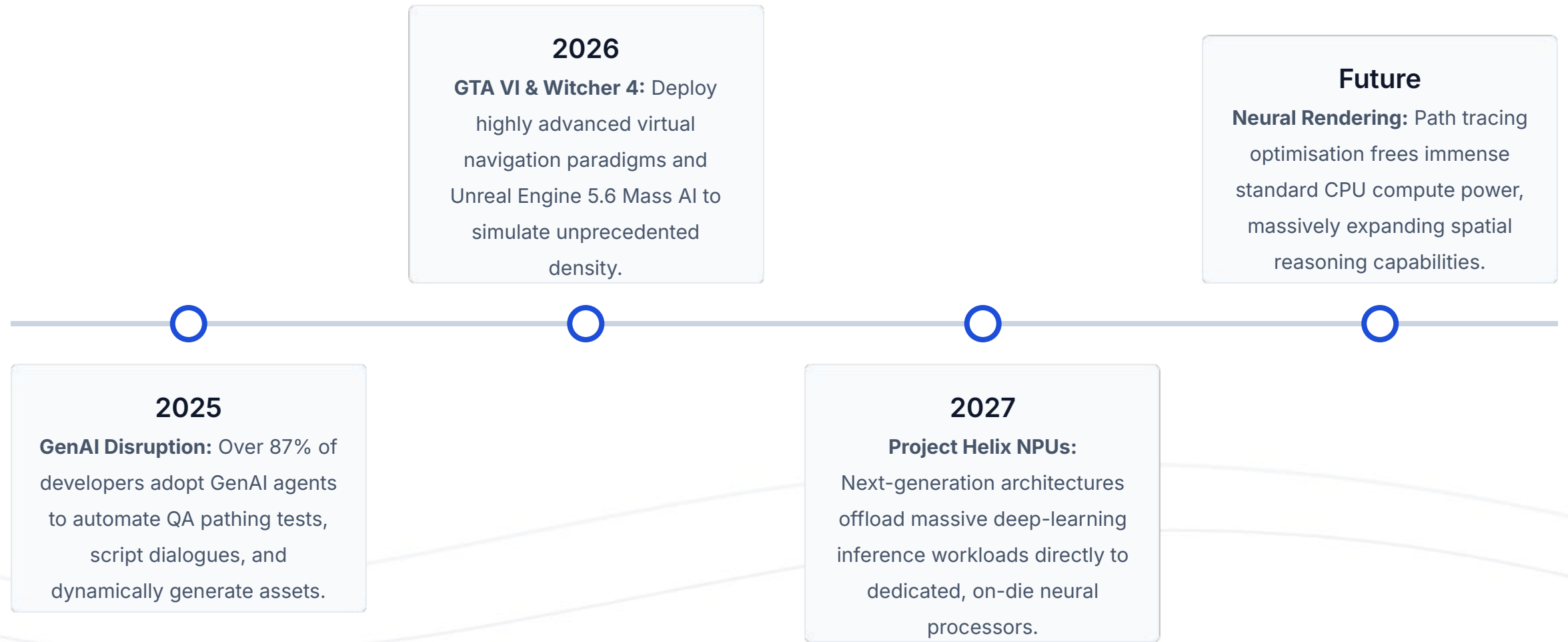
The Foundational Era (1978-2000)



Cognitive Leaps (2004-2015)



The Hardware Symbiosis (2025-2027)



Glossary of Key Terms

- **Heuristic Function** A mathematical estimator used in informed algorithms (like A*) to gauge the distance from a current node to the goal, heavily optimising search times by pruning irrelevant paths. $h(n)$
- **Data-Oriented Design (DOD)** A software engineering paradigm prioritising how hardware physically reads memory. Organises data in contiguous, homogeneous arrays (Struct of Arrays) to eliminate CPU cache misses.
- **Entity Component System (ECS)** The industry-standard architectural framework underpinning DOD. Entities are merely IDs, Components are pure data arrays, and Systems are isolated functions that iterate linearly over the data.
- **Goal-Oriented Action Planning (GOAP)** An AI decision-making architecture where agents are given a goal and dynamically search an abstract pool of available actions (using preconditions and effects) to formulate a real-time plan.
- **Navigation Mesh (NavMesh)** A spatial graph representing walkable terrain as a continuous network of convex polygons. Drastically reduces node counts and memory overhead compared to traditional grid-based arrays.
- **Object-Oriented Programming (OOP)** A traditional programming paradigm encapsulating state and behavior together in classes. Often causes performance-killing memory fragmentation when scaling up to simulate massive entity counts.

Questions?